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TECTONIC IMPLICATIONS OF PALEOMAGNETIC DATA FROM SILLS AND DYKES IN THE SVERDRUP BASIN, CANADIAN ARCTIC

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Abstract. Mesozoic intrusives, 71 sills and 12 dykes, from different structures in the folded eastern Sverdrup Basin contain stable, well-grouped paleomagnetic directions of both normal and reverse polarity. The mean direction is 274°, +79°, k=173, α_{95}=3.7°, with a corresponding pole position at 157°, 67°N, k=61, A_{95}=6.6°. The paleomagnetic fold test shows that the major deformation, involving tilting of strata around strike, postdated the emplacement of sills and dykes. Minor preintrusion deformation may also have occurred, but, more importantly, an offset of the paleopole from the Cretaceous section of the polar wander path permits a postintrusion anti-clockwise rotation of eastern Sverdrup Basin of about 15+/-20° with respect to stable North America. If the intrusives are Cretaceous, the degree of rotation is close to that (11°) estimated for Greenland during the opening of the Labrador Sea. No positive evidence has been found to support models of distributed shear with block rotations to accommodate the contentious northeast translation of Greenland along the Nares Strait.

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INTRODUCTION

Late Cretaceous and Tertiary plate tectonic activity in the North Atlantic, extending rifting into Labrador Sea and Baffin Bay, separated Greenland from the North American continent. The different stages, established from magnetic stripes [e.g. Srivastava, 1985], can be broadly correlated [Miall, 1984] with recognized episodes of deformation within the Sverdrup Basin and adjoining areas (Figure 1) during the Eurekan orogeny [Balkwill, 1978]. However, geophysical interpretations of large-scale strike-slip motion along the Nares Strait appear to contradict geological evidence [Dawes and Kerr, 1982]. To solve this problem, models have been proposed that involve distributed sinistral shear with strike-slip motion on faults [Miall, 1983] and block rotations around steep axes [Wynne et al., 1983].

Paleomagnetism is a powerful tool, complementing traditional structural analysis, for studying deformation. This paper presents a study of sills and dykes intruded into Triassic sediments and aims to determine the style of deformation during the Eurekan orogeny, in particular, the degree and extent of block rotations. The work is also relevant to the debate concerning the relation between magmatism, basin development, and rifting in the Arctic Ocean.

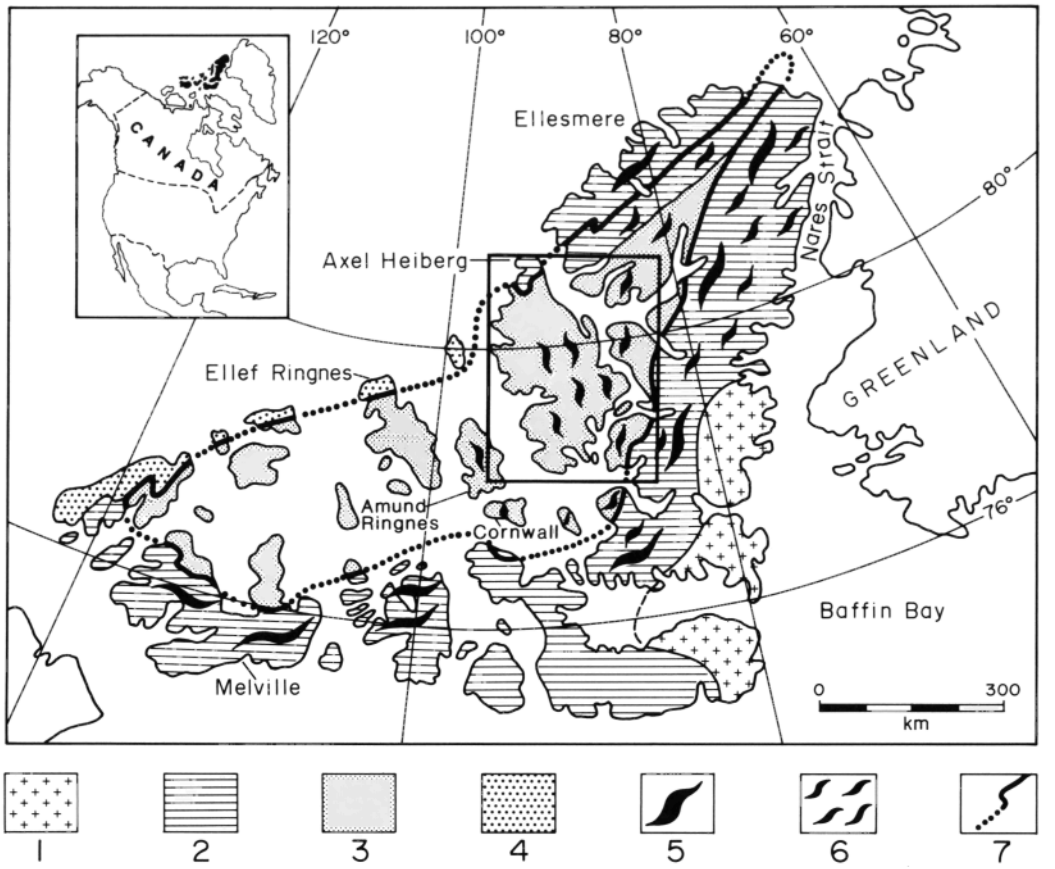


Fig. 1. Major stratigraphic and structural units in the Canadian Arctic Archipelago. Legend: 1, Precambrian; 2, Paleozoic; 3, Mesozoic; 4, Cenozoic; 5, approximate trend of Ellesmerian (Devonian); 6, Eurekan structures (Late Cretaceous-Tertiary); 7, Sverdrup Basin boundary; rectangle, see Figure 2.

GEOLOGICAL BACKGROUND

The rocks of the Sverdrup Basin, unconformably overlying Lower Paleozoic strata (Figure 1), comprise Lower Carboniferous evaporite followed by 13,000 M of Mesozoic to Lower Tertiary, mainly marine, clastics [Thorsteinsson and Tozer, 1970; Balkwill 1978]. In addition, tholeiitic basalts [Blackadar, 1964; Fricker, 1963; Williamson and Van Wagoner, 1985] appear either as lava flows or as sills and dykes. Early phases of effusive activity are recorded by the Upper Carboniferous Audhild and Lower Permian Esayoo basalts on the northwestern edge of the Basin on Axel Heiberg and Ellesmere islands. The main, Cretaceous phases [Osadetz and Ricketts, 1985] occurred between 130 to 90 Ma in the (1) Valangian to Aptian (Isachsen Formation), affecting the largest area, from Ellef Ringnes Island in the west to northwestern Ellesmere Island in the east; (2) late Albian to early Cennomanian (Hassel Formation), recorded by lava flows on Amund Ringnes Island [Balkwill, 1983]; and (3) Turonian (Strand Fiord Formation), restricted to thick flows on western Axel Heiberg Island.

The intrusive rocks are common and well exposed in the eastern part of the Basin. A few dykes on Melville Island, and evidence from drill cores and magnetic surveys [Balkwill and Fox, 1982], indicate that intrusives also occur in the western part. The dykes, rarely exceeding 10 km in length, are encountered in N-NNW trending swarms on northern Axel Heiberg and northwestern Ellesmere islands on either side of fault planes, although no dyke is observed to cross a fault. Dykes

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are also common, but lack regional trend, on southwestern Axel Heiberg, on Amund Ringnes, and on Cornwall islands. With the possible exception of intrusives near the Permian flows, the sills and dykes are considered by Thorsteinsson and Tozer (op. cit.) to be the intrusive equivalents of Cretaceous effusive activity. However, K-Ar ages [Larochelle et al., 1965; Balkwill, 1978, 1983; Balkwill and Haimila, 1978] for the sills range from 224 to 90 Ma, leading Balkwill to suggest that intermittent Mesozoic activity was connected with basin subsidence.

The major deformation of the Sverdrup Basin occurred during the Late Cretaceous to Tertiary Eurekan orogeny [Balkwill, 1978; Miall, 1984], although earlier tectonic influences include the Early Permian Melvillian disturbance, associated with the Esayoo basalt, and the rise of diapirs from Late Carboniferous evaporite, probably beginning in the Triassic and due to sediment loading [Balkwill, 1978]. The Eurekan orogeny is the result of the separation of Greenland from the North American continent. Because seafloor spreading began to the south, in the Labrador Sea, areas to the north of Greenland experienced compression. The first stage of separation, an 11° anticlockwise rotation of Greenland between 78 and 55 Ma around a pole at 95°W and 76°N (P in Figure 9; Srivastava [1985]), coincides with the development of broad basins and arches in Campanian to Paleocene. During the second stage (55-35 Ma), the rotation pole changed and Greenland moved to the northeast, creating Baffin Bay, and finally to the northwest in Eocene-Early Oligocene, resulting in the main phase of the Eurekan orogeny with regional compression in the eastern part of the Basin. Although the extent of the northeast translation of Greenland, assumed to have occurred as strike-slip motion on the Nares Strait, is still highly controversial [Dawes and Kerr, 1982], a solution may be provided by distributed sinistral shear on Ellesmere Island with strike-slip motion on faults [Miall, 1983] and block rotations (Wynne et al. [1983]; see also Figure 9). The mechanism would be similar to processes affecting the Cordilleras, where anomalous paleomagnetic poles have been explained by clockwise-rotated microplates linked to dextral shear between the North American and Pacific plates [e.g., Nicholson et al., 1986].

DESCRIPTION OF SAMPLING LOCALITIES

The paleomagnetic sampling concentrated upon eastern Axel Heiberg and northwestern Ellesmere islands (Figure 2), which formed the depocenter of the Sverdrup Basin in the early Mesozoic, and where the structural style is less complicated than further west.

The Skarre Fiord Syncline (sections I-III; Figure 3) is bounded to the east by the west dipping Stolz Thrust Fault of unknown displacement. The trend of both syncline axis and thrust edge changes southward from northwest-southeast to north-south. The eastern limb of the Syncline contains the thickest Triassic sequence in the Sverdrup Basin, up to 6000 M of siltstone, black shale, and sandstone. Section IV of east dipping sills was sampled on an anticline further southwest. In the Blue Mountains (sections V and VI) the lithology is similar to that of the Skarre Fiord Syncline, except that layers of quartzitic sandstone are interbedded in Middle-Upper Triassic dark shale. Here too the structure consists of a bent thrust edge and syncline axis, although the general trend is northeast-southwest to east-west. A syncline, cut by three dykes, also occurs along the hanging wall side of the Van Hauen Thrust Fault (Figure 4). Bjarnasson Island contains a thinner, generally coarser-grained basin marginal section; unfortunately, only three sills in Middle Triassic shale could be sampled due to severe weather conditions.

The sills preferentially appear in Middle-Upper Triassic black shale, often forming prominent ridges that can be followed for tens of kilometers, and are completely absent in Jurassic and Cretaceous sediments in this part of the Sverdrup Basin. Field and air photo observations indicate that a particular sill generally can be followed along strike for no more than a few kilometers, after which it tapers out and another, slightly overlapping sill may appear immediately below or above. Thus the sills do not appear as large sheets, covering many square kilometers, but rather as "leaves" branching out from what must be an extensive, interconnected magma supply system. Locally, sills may climb by cutting the strata at a low angle or, as seen in excellent cliff exposures along Buchanan Lake immediately north of section II, by abruptly turning into a dyke

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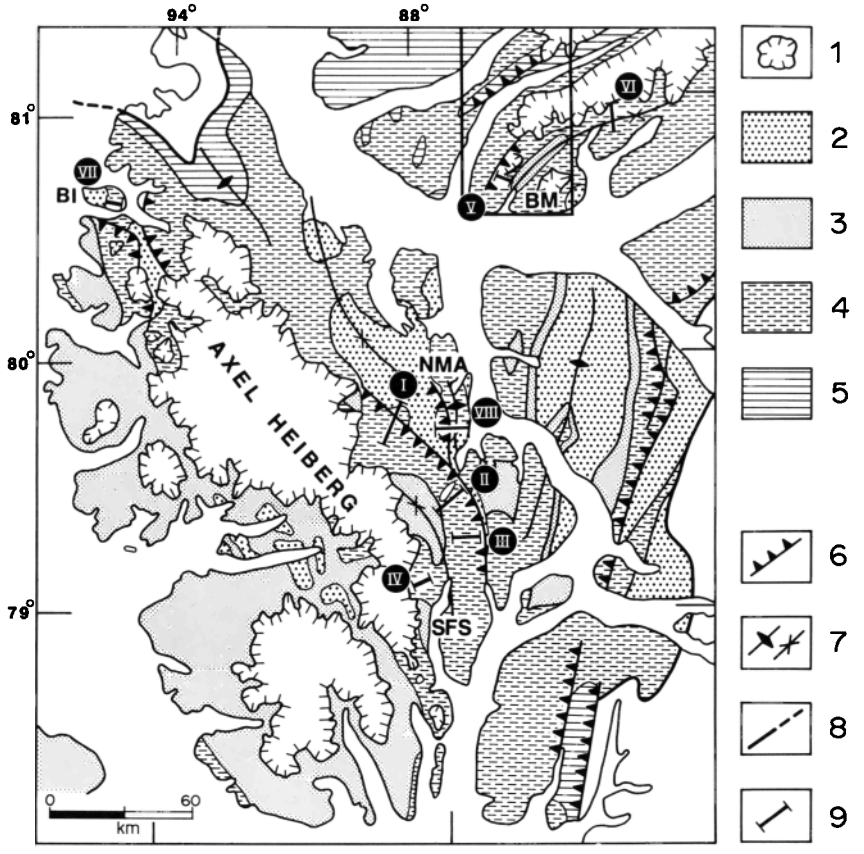


Fig. 2. Geological map of Axel Heiberg and northwest Ellesmere islands, from Geological Survey of Canada maps 1299A, 1391A, 1302A, 1305A, 1306A, 1308A-1311A. Legend: 1, ice field; 2, Tertiary and 3, Jurassic-Cretaceous clastics; 4, Triassic shale, siltstone, and sandstone; 5, Permo-Carboniferous limestone, evaporite, and clastics; 6, thrust fault, teeth on upthrust side; 7, anticlinal, synclinal axis; 8, Sverdrup Basin boundary, areas outside left blank; 9, sampled sill section; SFS, Skarre Fiord Syncline; NMA, North Mokka Anticline; BM, Blue Mountains; rectangle, see Figure 4.

cutting an overlying sill and then turning back into a sill again. The volume of magmatic material, as measured by the accumulated thickness of sills (Table 1), increases to the north on the Skarre Fiord Syncline and to the east in the Blue Mountains, presumably indicating proximity to the source.

The dykes sampled on northwestern Ellesmere Island (Figure 4) occur in folded Triassic sediments and appear to feed not only sills but also lava flows known to be Cretaceous [Thorsteinsson and Tozer, 1970]. On air photos the dykes can be traced for 2-9 km. Based on site observations, the strike (dip) of the dykes averages 155+/-10° (80°E-75°W). In hand sample the dykes are more weathered than the sills, presumably because the latter, exposed in steep cliffs, are more rapidly eroded. The occurrence of slickensides and fractures shows that dykes, as well as sills, have been deformed together with the sediments.

SAMPLING AND LABORATORY PROCEDURES

Five to eight hand samples, oriented using a sun compass, were taken at each site (Table 1). Where possible, samples were distributed from one chilled margin to the center of the intrusive. The strike and dip of the sedimentary strata (mean of several observations) were measured near the sampling site, as close to the contact as possible.

From each hand sample 2.5x2.5 cm specimen cores were drilled. One specimen

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per sill sample was thermally demagnetized using a Schonstedt Model STD-1 Thermal Specimen Demagnetizer, with its remanence measured on a Schonstedt Spinner Magnetometer Model SSM-1. Alternating field demagnetization was undertaken on dyke and selected sill specimens using a Schonstedt Model GSD-1 AC Geophysical Specimen Demagnetizer and a Digico Spinner Magnetometer. The visual identification of components was complemented by computer analysis, using either converging remagnetization circles [Bailey and Halls, 1984] or a modified version [Halls, 1986] of a Principal Component Analysis program [Kirschvink, 1980], wherein all possible fitted lines (planes) that show a mean angular deviation smaller than 5° (10°) are calculated. Curie points were measured in air with a Cahn RG Electrobalance and a 10^{5} A/m (5 kOe) magnetic field.

PALEOMAGNETIC RESULTS

Sills

The collection contains 451 hand samples from 71 sills in sections I-VII. In the interior of sills the intensity of natural remanent magnetization (NRM) is generally around 1 A/m (10^{-3} emu/cm). Toward the margin there is often a decrease in intensity accompanied by a zone of low unblocking temperatures and irregular demagnetization behavior, and 16 out of 20 discarded samples with mean angular deviation for lines [Kirschvink, 1980] larger than 5° were taken within the first few decimeters from the margin.

The sills can be divided into two groups based on their thermal demagnetization behavior. The first group (Figures 5a-c) contains a simple magnetization whose intensity decreases rapidly around 550°C, in addition to varying decay between 250° and 450°C. In specimens from the top of section II, alternating field demagnetization behavior and a single Curie point at 570°C indicate the presence of (pseudo-) single-domain magnetite, and the variation in unblocking temperatures is probably due to two discrete grain size assemblages. Specimens further down the section contain higher coercivity material and occasional maghemite. The direction of components residing in different parts of the blocking temperature spectrum may vary within 5°, exceptionally 10°, (Figure 5b). However, within a sill, there is no statistically significant difference between "low-" and "high-temperature" components, and these minor fluctuations may be due to secular variation of the ambient field during cooling of the intrusives. The high-temperature component is a stable endpoint, usually with mean angular deviation smaller than 3°.

Sills of the second group, indicated by an asterisk in Table 1, display a complex thermal behavior (Figure 5d), but alternating field demagnetization (Figure 5e) often yields stable endpoints similar to those from the first group. In addition, these specimens commonly contain large low-coercivity components that do not define a coherent direction of magnetization. The complex thermal behavior may be due to a self-reversal in material produced during the heating but does not affect the following discussion.

The mean of the stable endpoints obtained, the site direction, is usually well defined (Table 1); only two site directions, from sills 18 and 58, with axes of the 95% confidence ellipse larger than 20°, are not included in the subsequent analysis. The majority of sills are normally magnetized, but a few have reversed polarity. Sites 20, 83, 94, and 95 contain stable endpoints of both polarities but are listed according to the dominant polarity as it was not possible to determine which represented the primary remanence. The mixed polarity may be due to remagnetization or slow cooling in a reversing field, or to a process similar to the "self-reversal" encountered during thermal demagnetization. However, true reversals of the Earth's magnetic field have been recorded because reversed components of magnetization occur in material with widely differing coercivities and unblocking temperatures.

Dykes

The collection contains 86 hand samples from 16 sites in 12 dykes (Figure 4). The NRM intensity is generally around 1 A/m, and all specimens were demagnetized by alternating fields. Sites 110, 113, 114, and, in part, 111, 115, and 116 contain a simple magnetization with decay behavior typical of single-domain magnetite (Figure 5f). The remainder shows a multidomain behavior with large viscous overprints in a recent Earth's field (Figure 5g).

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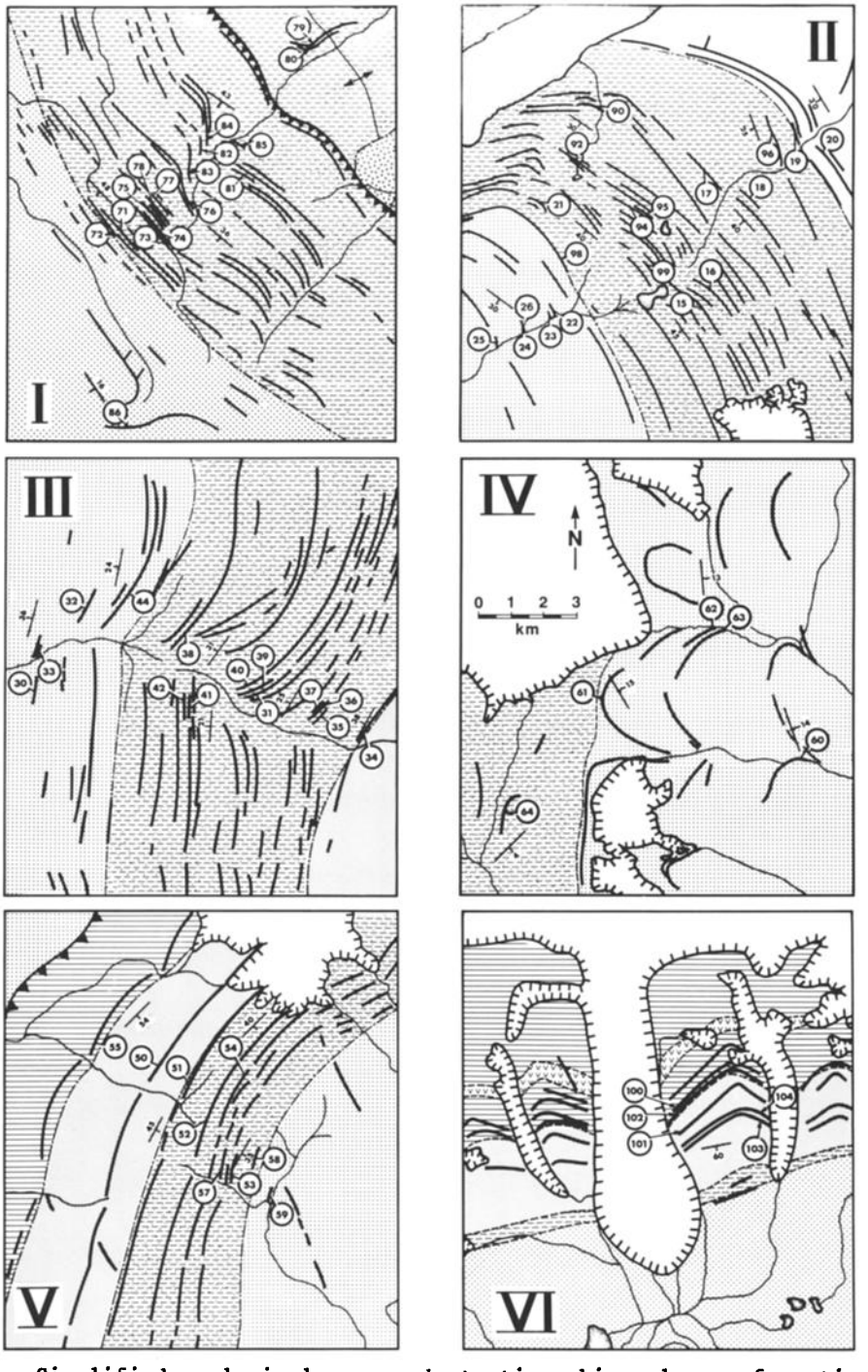


Fig. 3. Simplified geological maps and stratigraphic columns of sections I to VI, based on field evidence, air photos, and Geological Survey of Canada maps. Legend: 1, Tertiary clastics; 2, Upper Triassic sandstone; 3, Middle-Upper Triassic shale; 4, Lower Triassic sandstone; 5, Permian Esayoo basalt; 6, Permo-Carboniferous sediments; 7, thrust fault, teeth on upthrust side; 8, bedding near sampled site; 9, anticline; 10, shoreline; 11, ice field; 12, crest of sill; 13, site number and location; 14, sill with mixed or reversed polarity of magnetization.

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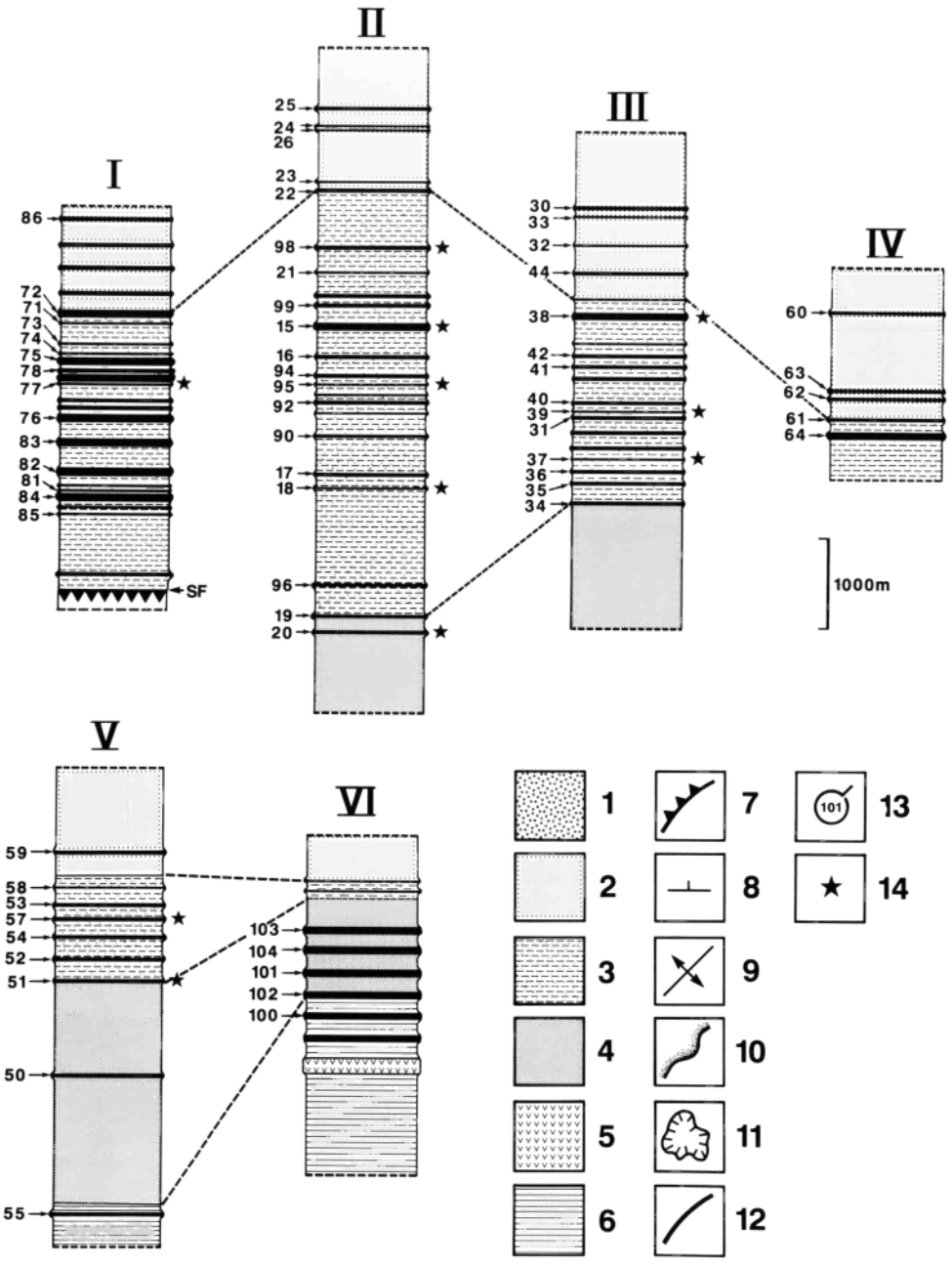


Fig. 3. (continued)

The site directions are listed in Table 1; five out of seven site directions from the Blue Mountain dykes (BMF) show unacceptably large errors due to scattered components that could not be removed. As an area mean direction (compare with Table 2) cannot be calculated based on only two sites, 106 and 110 with directions similar to those from other intrusives, these dykes have been excluded from further consideration, and the "Blue Mountains area" hereinafter refers to sill sections V and VI.

DISCUSSION OF RESULTS

Site directions, before and after unfolding of the sediments to the

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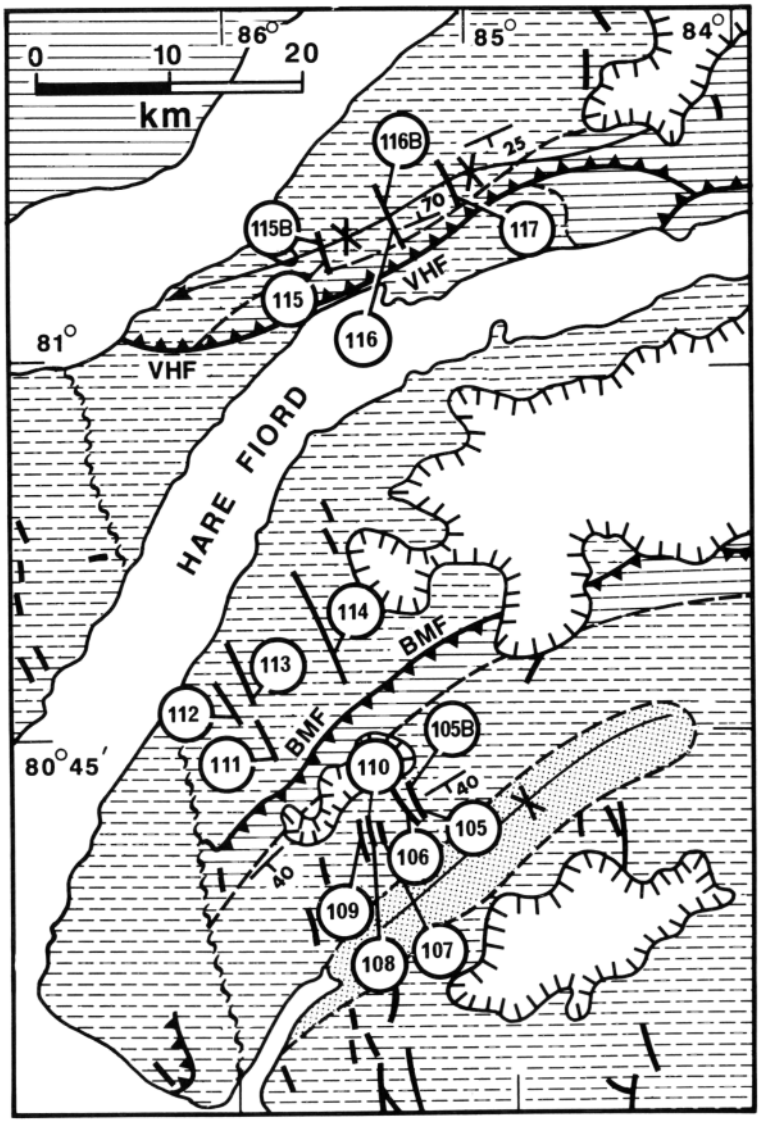


Fig. 4. Geological map of area surrounding Hare Fiord on northwest Ellesmere Island, based on Geological Survey of Canada maps 1309A and 1311A. Legend as in Figure 2, except that dykes are shown by heavy lines; BMF, the Blue Mountains Thrust Fault; VHF, the Van Hauen Thrust Fault; squiggles, normal fault; encircled numbers, sampling sites; conventional symbols indicate strike and dip of strata. The Cretaceous sequence southeast of BMF includes a 15-m lava flow.

horizontal around strike, are compared in Figure 6 with directions that yield poles that fall on the Mesozoic-to-present section of the apparent polar wander path for stable North America [Irving and Irving, 1982]. Following Steiner [1983] and May and Butler [1986], the relatively uncertain mid-to-late Jurassic section of the path is assumed to be constrained to a latitudinal distribution between the Triassic-early Jurassic and the Cretaceous clusters. The scattered Jurassic paleopoles are individually plotted in Figure 9, where the mean pole position for the Sverdrup Basin is compared with other North American results. Although the Cretaceous section is based on only eight determinations, of which four define the Early Cretaceous, they are all obtained from igneous rocks and are closely grouped

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TABLE 1. Site Directions from Sills (15-104) and Dykes (105-117) in Eastern Sverdrup Basin

Location					Paleomagnetic data				
		Width	Bedding		In situ		Unfolded		
Section	Site	m	Strike	Dip	D	I	D	I	N / k / α_{95}
I	86*	13	146	+16	327	+79	271	+71	5/187/ 5.6
I	72	30	134	+49	30	+55	253	+73	6/ 34/11.6
I	71	8	134	+49	30	+53	259	+75	6/ 98/ 6.8
I	73*	111	34	+44	8	+59	270	+65	5/ 18/18.4
I	74	5	130	+44	24	+57	256	+75	5/489/ 3.5
I	75	70	135	+44	32	+56	259	+77	4+1/860/ 3-3
I	78	15	134	+42	40	+57	238	+81	4/599/ 3.8
I	77	25	127	+41	201	-39	162	-75	5/ 21/17.3
I	76	117	131	+36	21	+62	266	+77	5/ 98/ 7.8
I	83**	80	131	+37	25	+53	305	+80	5/ 29/14.6
I	82*	45	131	+40	47	+64	210	+76	0+5/114/ 8-9
I	81	2	131	+48	18	+61	249	+67	5/210/ 5.3
I	84	76	123	+43	32	+71	214	+66	5/139/ 6.5
I	85	3	137	+37	88	+82	217	+59	5/ 61/ 8.6
I	79	26	179	+ 8	227	+80	245	+73	4/104/ 9.1
I	80	22+	179	+ 8	315	+81	294	+74	5/ 48/11.2
II	25	30	155	+25	25	+76	278	+73	7/262/ 3.7
II	24	8	155	+33	62	+68	251	+79	5/ 71/ 9.1
II	26	3	155	+33	62	+70	250	+77	6/435/ 3.2
II	23	6	144	+34	44	+62	270	+82	5/647/ 3.0
II	22	30	138	+37	56	+61	203	+81	7/525/ 2.6
II	98*	10+	135	+40	209	-40	171	-75	3+1/76/11-13
II	21	8	126	+42	34	+61	220	+77	6/ 57/ 8.9
II	99	28	140	+44	36	+65	247	+70	4/413/ 4.5
II	15	45	142	+45	236	-47	359	-87	7/ 59/ 7.9
II	16	20	142	+45	58	+52	205	+82	6/ 82/ 7.4
II	94**	23	132	+42	32	+67	234	+70	4+3/ 23/11.8
II	95**	7	132	+42	210	-64	60	-73	2+5/54/10-13
II	92	30	125	+40	27	+63	231	+76	6/117/ 6.2
II	90	15	129	+25	44	+70	200	+85	5/159/ 6.1
II	17	15	139	+43	33	+59	262	+75	10/ 84/ 5.3
II	18*	5	139	+38	232	-71	46	-71	2+4/13/23-35
II	96	30	159	+25	25	+72	296	+73	4/ 18/17.6
II	19	23	149	+35	50	+70	251	+75	9/257/ 3.2
II	20**	15	149	+31	235	-65	75	-84	6/120/ 6.1
III	30*	10+	195	+36	102	+71	288	+73	8/299/ 3.2
III	33	7	195	+36	102	+72	288	+72	7/188/ 4.4
III	32	12	195	+36	90	+72	299	+71	8/ 41/ 8.8
III	44	22	189	+24	97	+74	283	+82	6/416/ 3.3
III	38*	30+	215	+31	298	-65	151	-83	7/ 70/ 7.3
III	42	17	187	+21	109	+75	251	+83	5/905/ 2.5
III	41	18	187	+21	107	+76	259	+82	6/746/ 2.5
III	40	15	204	+26	129	+76	278	+77	5/138/ 6.5
III	39*	10	204	+26	293	-75	115	-79	5/ 81/ 8.6
III	31	15	204	+26	117	+71	286	+83	6/645/ 2.6
III	37*	9	212	+26	341	-83	110	-69	7/297/ 3.5
III	36	30	212	+26	156	+78	279	+73	7/275/ 3.6
III	35	30	212	+26	167	+77	273	+71	6/100/ 6.7
III	34	14	208	+38	127	+67	285	+74	6/722/ 2.5
IV	60	18	340	+14	266	+74	322	+85	8/ 89/ 5.9
IV	63*	15+	10	+15	283	+72	297	+87	6/125/ 6.0

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TABLE 1. (continued)

Location					Paleomagnetic data				
		Width	Bedding		In situ		Unfolded		
Section	Site	m	Strike	Dip	D	I	D	I	N / k / α_{95}
IV	62	10	355	+13	284	+69	309	+80	7/1341/1.6
IV	61	22	329	+15	250	+70	278	+84	4/190/ 4.2
IV	64	35	45	+ 4	290	+80	275	+83	7/603/ 2.5
V	59*	14	30	+45	283	+41	222	+77	6/ 24/13.8
V	58	5	26	+31	258	+81	129	+66	2+3/15/24-25
V	53	18	25	+38	296	+58	110	+84	7/277/ 3.3
V	57*	15	25	+38	110	-19	106	-57	4+1/181/ 6-7
V	54	23	48	+40	290	+52	211	+72	5/284/ 4.5
V	52	25	29	+45	285	+45	204	+80	10/ 76/ 5.6
V	51*	15	41	+47	121	-30	96	-75	2+3/586/ 4-6
V	50	25	44	+45	301	+39	251	+79	6+3/128/ 5-6
V	55	18	48	+54	311	+35	236	+84	7/435/ 2.9
VI	103*	56	84	+57	333	+25	282	+70	6/182/ 4.2
VI	104	85	84	+57	343	+33	261	+81	8/180/ 3.7
VI	101	35	76	+64	345	+23	329	+87	8/107/ 4.8
VI	102*	90	80	+64	345	+25	271	+85	7/ 81/ 5.9
VI	100	34	81	+60	333	+27	267	+74	3+4/173/ 5-7
VII	89	18	211	+40	139	+58	255	+77	5/149/ 5.1
VII	88	52	16	+31	146	+58	219	+80	5/ 84/ 6.8
VII	87	82	13	+45	153	+46	227	+69	5/ 16/15.6
BMF	105	20	50	+29	332	+57	22	+83	0+5/119/12-22
BMF	105B	20	51	+37	330	+41	352	+77	0+4/20/48-85
BMF	106	6+	50	+40	338	+64	112	+73	3+2/ 82/9-11
BMF	107	8	54	+39	170	-59	273	-73	2+3/ 8/34-53
BMF	108	15	42	+36	333	+54	51	+78	3+2/17/22-24
BMF	109	8	45	+38	326	+54	66	+83	1+4/10/37-56
BMF	110	25	41	+41	310	+42	305	+83	5/143/ 6.4
HF	111	19	189	+11	151	-86	112	-76	4+2/250/ 4-5
HF	112	6	152	+15	45	+87	246	+78	6/ 64/ 8.4
HF	113	10+	160	+15	120	+82	218	+78	4/361/ 4.8
HF	114	13	166	+15	232	+84	249	+69	6/189/ 4.9
VHF	115	14	239	+71	335	-27	116	-80	4+2/325/ 4-5
VHF	115B	14	76	+24	162	-65	105	-88	2+4/187/ 6-9
VHF	116	12	239	+68	156	+44	316	+67	4+2/39/12-13
VHF	116B	17	62	+29	328	+48	320	+77	4+2/ 61/9-11
VHF	117	15	240	+52	157	+44	291	+82	4+2/ 85/ 8-9

D, declination; I, inclination; N, number of stable endpoints (plus remagnetization circles; Bailey and Halls [1984]); k, Fisher's precision parameter; α_{95}, radius (minimum and maximum axes) of the 95% confidence cone (ellipse).

*Sills with complex demagnetization behavior.

**Sills with stable endpoints of both polarities.

and consistent. One Arctic study, the Isachsen lavas on Ellef Ringnes Island [Larochelle and Black, 1963], is included in the calculation of the path.

As a result of unfolding, the site directions, initially scattered and unrelated to the path, move close to the Cretaceous section and do not conform with early Jurassic-Triassic directions listed by Irving and Irving. Note, further, that there is no paleomagnetic evidence that the magnetization age of sills increases with stratigraphic depth in the thick sections on eastern Axel Heiberg Island as

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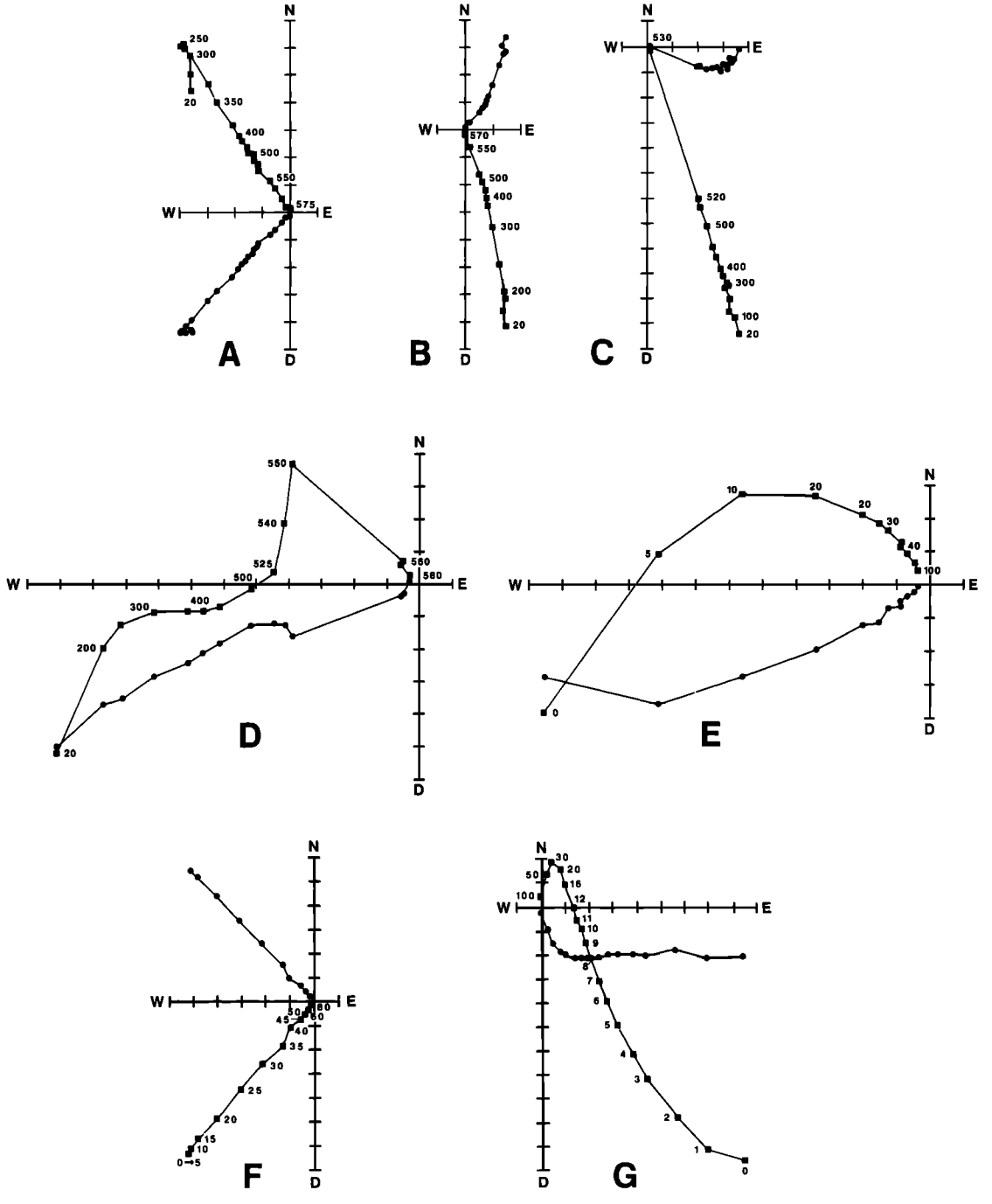


Fig. 5. Thermal demagnetization of sill specimens from (a) reversely magnetized site 15, and (b), (c) normal sites 76 and 31; thermal (d) and alternating field (e) demagnetization of specimens from the same hand sample in site 33; and alternating field demagnetization of dyke specimens from sites 110 (f) and 115 (g). Orthogonal plots with demagnetization step in ^{o}C or mT. Solid circles and squares represent projections of NRM vector tip onto horizontal, vertical planes. Scale divisions in 0.4 (Figures 5a and 5b), 0.2 (Figures 5c, 5f and 5g), and 0.1 (Figures 5d and 5e) A/m.

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TABLE 2. Section and Mean Directions for Eastern Sverdrup Basin

Location			In situ			Unfolded		
Section	°W	°N	D	I	N / k / α_{95}	D	I	N / k / α_{95}
I	88.50	79.68	25	+65	16/ 23/ 7.8	254	+75	16/ 56/ 5.0
II	87.42	79.38	43	+64	18/ 57/ 4.6	251	+80	18/ 85/ 3.8
III	87.33	79.20	117	+75	14/119/ 3.7	285	+77	14/169/ 3.1
IV	88.33	79.13	273	+74	5/160/ 6.1	296	+84	5/571/ 3.2
V	85.78	80.63	295	+40	8/ 32/ 9.9	247	+79	8/ 39/ 9.0
VI	83.08	80.87	340	+27	5/143/ 6.4	275	+80	5/105/ 7.5
VII	95.13	80.65	147	+54	3/100/12.4	233	+76	3/125/11.1
VH	85.33	81.12	158	+70	5/ 3/52.3	309	+79	5/100/ 7.7
HF	85.83	80.77	131	+89	4/158/ 7.3	252	+77	4/ 97/ 9.4
NMA	[Jackson & Halls, 1985]					286	+72	12/137/ 4-5
					mean 1[a]:	267	+79	10/187/ 3.5
					mean 2[b]:	277	+79	10/214/ 3.3
					mean 3[c]:	274	+79	10/173/ 3.7
VGP[d]:	157°W	67°N	N=10	K=61	A_{95}=6.6			

D, declination; I, inclination; N, number of site directions; k, Fisher's precision parameter; α_{95}, radius of the 95% confidence cone.

[a]Based on unfolded section directions equally weighted.

[b]As before, with 30° clockwise oroclinal rotation of sections I, II and V.

[c]Based on mean for sections I-III and V-VI and NMA weighted by 2, and site direction IV, VII, V and HP weighted by 1.

[d]Virtual geomagnetic pole corresponding to mean 3.

predicted by Balkwill's [1978, 1983] model of intermittent intrusive activity connected with periods of Mesozoic basin subsidence. For example, a mean direction based on sills 20-16 in section II (256°, +81°, α_{95}=6°) is not significantly different at the 95% confidence level [Watson, 1956] from a mean [247°, +78°, α_{95}=5°) of sills 15-25. We therefore believe it reasonable to assume that there is not a major spread in the magnetization age of intrusives from separate areas and that differences in magnetic directions can be interpreted in terms of deformation.

Although the absolute age of the sills and dykes cannot be determined by paleomagnetic means (compare with below), we speculate that without a clear link between extended Mesozoic basin subsidence and magmatism the intrusives should instead be correlated with the Cretaceous lava flows [Thorsteinsson and Tozer, 1970]. The absence of apparent polar wander between 130 and 80 Ma precludes a specific correlation with the three phases of Cretaceous extrusive activity, with the exception of the reversely magnetized sills and dykes which must predate the long period of normal polarity between 115 and 75 Ma [LaBreque et al., 1977; Larson and Hilde, 1975] and therefore would be equivalent to the Valangian-to-Aptian lava flows of the Isachsen Formation.

The paleomagnetic "fold test" [Graham, 1949; McDonald, 1980] requires that a stable remanence predates and survives deformation. In a deformed terrain it is therefore possible to compare the grouping of magnetic directions before and after structural unfolding. The application of the fold test to the results obtained in this study is illustrated by Figure 7, using, for clarity, the mean direction for each section. A simple unfolding of the sediments to the horizontal around strike brings the scattered in situ directions close together, increasing the precision parameter (k) from 5 to 184, which is statistically significant even at the 99% confidence level [McElhinny, 1964]. Therefore the major part of the deformation, involving rotation around presently observed strike, postdates the magnetization and thus the emplacement of the sills and dykes.

A closer consideration of the data (Figure 8) suggests that the best grouping of results is not obtained by complete unfolding of the beds to the horizontal.

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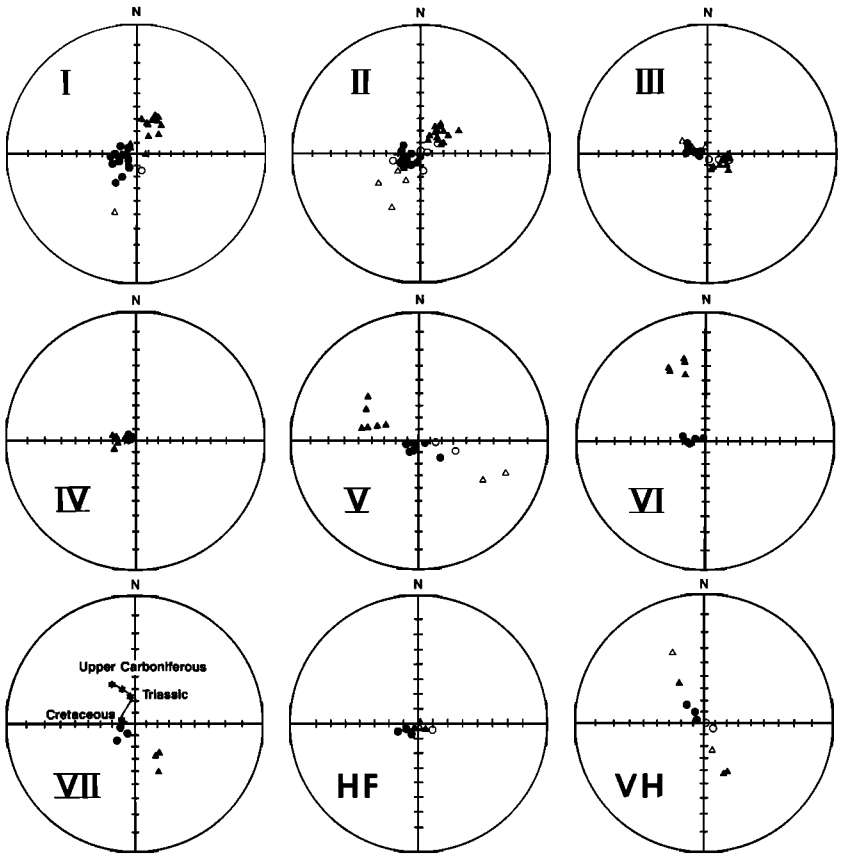


Fig. 6. Site directions for sills from sections I to VII, and dykes from Hare Fiord (HF) and Van Hauen (VH) areas, before (solid triangles) and after (solid circles) correction for tilt of adjacent sedimentary rocks. Lower left: Directions (asterisks) that would fall on the Mesozoic-to-present section of the apparent polar wander path for stable North America superimposed on results from section VII. Wulff equal angle stereonets, solid triangles and circles, lower and open triangles and circles, upper hemisphere.

First, within the western hanging wall side of the Stolz Thrust Fault in section I (Figure 3), site directions 81, 82, 84, and 85 move from an in situ position in the northeast quadrant of the stereonet to positions beyond that of the main cluster of site directions as indicated by the nonoverlapping circles of 95% confidence. The grouping could be improved if it were assumed that at the time of intrusion the beds near the (present) edge of the thrust were already tilted relative to the remainder. Such a tilt could be due to a normal fault in support of the suggestion [van Berkel et al., 1983] that Eurekan thrusts were predetermined by older structures. Secondly, section directions I to VI show a "crossover" of paths followed during unfolding, which may indicate a small preintrusion tilt around present strike of the sedimentary beds. It is also possible to reduce the scatter within the eastern limb of the Skarre Fiord Syncline and the Blue Mountains by oroclinal bending, in this case implying that the present changes in axial trends are postintrusion features. Although the paleomagnetic fold test cannot distinguish between these alternatives, it should be noted that a preintrusion deformation, due to evaporite dispirism, has been documented from the North Mokka Anticline on eastern Axel Heiberg Island [Jackson and Halls, 1985]. Furthermore, dyke swarms continue across the Blue Mountains Thrust without offset or change in strike (Figure

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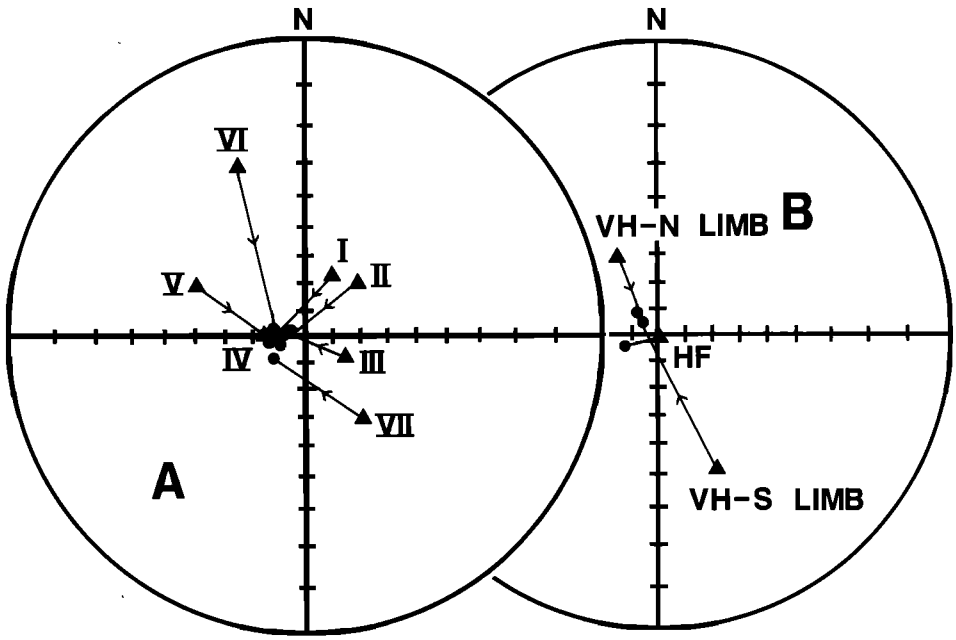


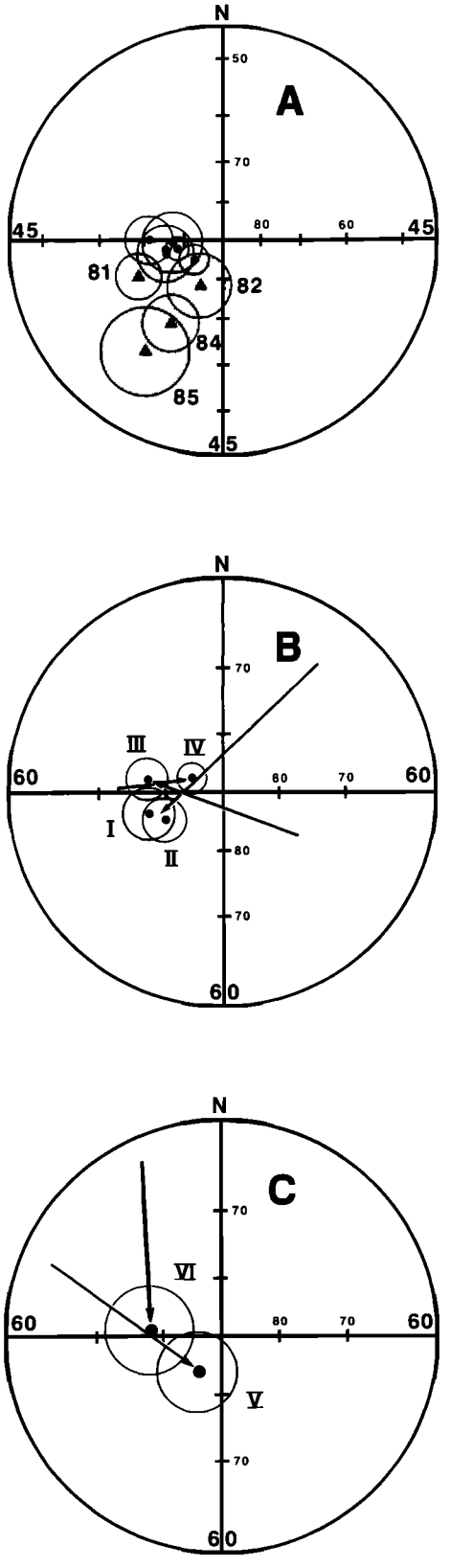
Fig. 7. Mean of site directions for each area of (a) sills and (b) dykes, before (solid triangles) and after (solid circles) structural correction. If a sill or dyke is reversely magnetized, the antiparallel direction was used to calculate the mean. Wulff equal angle stereonet, where all directions are plotted on lower hemisphere.

4), suggesting that any transcurrent component of motion was minimal along these faults.

The total mean direction for the Sverdrup Basin intrusives (274°, $+79^{o}$, $\alpha_{95}=3.7^{o}$; mean 3 in Table 2) was calculated assuming a small preintrusion tilt within the Blue Mountains and the eastern limb of the Skarre Fiord Syncline with rotation paths due to unfolding approximating great circles [Bailey and Halls, 1984]. The mean direction for these two areas and for the North Mokka Anticline, based on a large number of sites and by unfolding structures with substantial internal differences in strike and dip, were weighted by 2 and section IV, Bjarnasson Island, the Van Hauen and Hare Fiord directions by 1. Alternative mean directions, based either on complete unfolding or on postintrusion bending of synclines, are also listed to show that reasonable variations in internal deformation would not drastically alter the results. However, by using a mean direction which assumes a preintrusion tilt for comparison with the apparent polar wander path, we will avoid overestimating possible postintrusion deformation.

The virtual geomagnetic pole position (VGP) that corresponds to the mean direction (Figure 9) lies slightly to the east of the Cretaceous section of the apparent polar wander path for North America [Irving and Irving, 1982]. The overlapping circles of confidence show that this offset is not significant at the 95% confidence level if the intrusives are Cretaceous, while displacement of the eastern Sverdrup Basin with respect to the North American plate is required if the magnetization age is Jurassic or Triassic. Wynnne et al. [1983], to explain a $36+/-8^{o}$ offset from the apparent polar wander path of the VGP from the Permian Esayoo basalt on northern Ellesmere Island, suggest that the Basin may have experienced two phases of rotational movement: (1) anticlockwise rotation of the eastern Sverdrup Basin with Greenland during the initial opening of the Labrador Sea between 78 and 55 Ma around a pole at 95°W and 76°N (Srivastava [1985]; P in Figure 9) and (2) additional local block rotation around vertical axes in response to subsequent sinistral shear

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as Greenland moved to the northeast creating Baffin Bay. To test whether such rotations are compatible with our result, the Sverdrup Basin VGP can be rotated using either P or a vertical axis because of the close proximity of P to the sampled area. The conversion of directions into paleopoles decreases the precision (k) to the extent that instead of the standard Watson [1956] method, the approximate test developed by McFadden and Lowes [1981] for mean directions with significantly different precisions had to be used. A 15° clockwise rotation of the Sverdrup Basin VGP around P brings it closest to the Cretaceous paleopoles, as given by the maximum combined vector sum, with an error of $+/-20^{\circ}$ within which the two means are not significantly different at the 95% confidence level. With increasing rotation around P, the VGP moves to the north of the apparent polar wander path and cannot be rotated to agree with early Jurassic or older sections.

Thus if the age of the intrusives is Cretaceous as seems likely, the paleomagnetic results can readily accommodate, but do not require, a 15° anticlockwise rotation of the eastern Sverdrup Basin at any stage in the Eurekan orogeny. Although the precise timing cannot be established from the paleomagnetic results, it is suggestive that most of the estimated rotation (11° of 15°) can be accounted for by the initial opening phase of the Labrador Sea. Such a rotation need not be interpreted as a rigid block movement of the entire basin but may represent an average of internal deformations occurring during a Campanian-Paleocene narrowing of the basin due to the rotation of Greenland [Peirce, 1982]. If this is the case, there is no positive evidence that any significant rotation has occurred during the

Fig. 8. (a) During unfolding, the site directions from section I move from the northeast to the southwest quadrant; directions from sills closest to the thrust edge (solid triangles) move beyond the main cluster. Arrows indicate approximate path of section directions from (b) the Skarre Fiord Syncline and (c) the Blue Mountains during unfolding. Enlarged central portion of Wulff equal angle stereonets, perimeter equals 45° (Figure 8a) or 60° (Figures 8b and 8c); circles at 95% confidence level.

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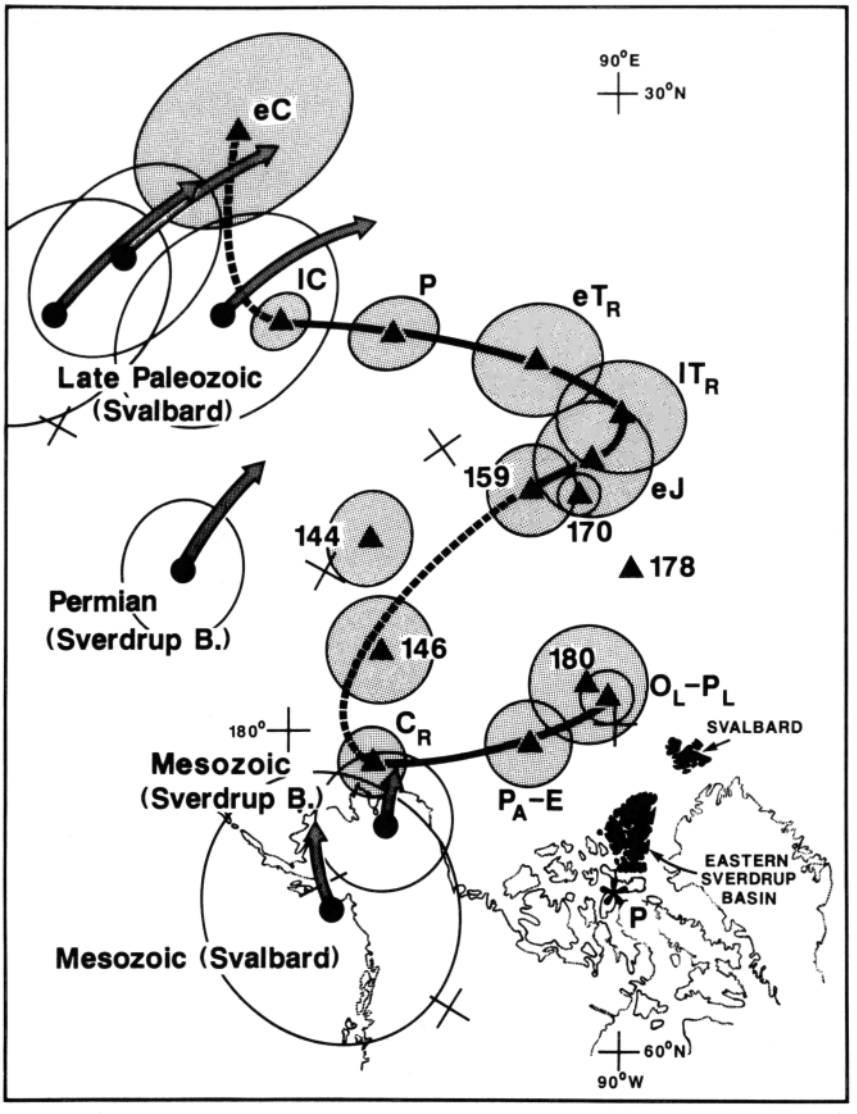


Fig. 9. Comparison of paleopoles (solid circles) from eastern Sverdrup Basin [this study; Wynne et al., 1983] and Svalbard [Vincenz and Jelenska, 1985; Torsvik et al., 1985; Watts, 1985] with the apparent polar wander path for North America (solid triangles), recalculated from Irving and Irving [1982] and including an early Carboniferous pole from Irving and Strong [1984]. Svalbard and its paleopoles is returned to 55-Ma position (anomaly 25) relative to North America; Greenland is returned to prerift position at 78 Ma (anomaly 33). Arrows indicate the change in paleopoles with a 15° anticlockwise rotation around P of eastern Sverdrup Basin and Svalbard with Greenland between 78 and 55 Ma. C, Carboniferous (e, early, and l, late); P, Permian; T_R, Triassic; J, Jurassic; C_R, Cretaceous; P_A-E, Paleocene-Eocene; O_L-P_L, Oligocene-Pliocene; age in Ma beside scattered Jurassic poles; asterisk, rotation axis P; circles of 95% confidence distorted at lower latitudes due to equal area projection.

subsequent compressional phase either by folding or by block rotations involving sinistral shear.

There is a striking similarity (Figure 9) between the results from the eastern Sverdrup Basin and those, albeit scattered, from Mesozoic intrusives and Permo-Carboniferous sediments on Svalbard [Torsvik et al., 1985; Vincenz and Jelenska, 1985; Watts, 1985]. The

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easterly displacement of Svalbard VGPs from the apparent polar wander path raises the possibility that the rotation of Greenland prior to 55 Ma affected Svalbard as well. The post-55 Ma deformation of Svalbard is unlikely to have produced the observed rotation because it is associated with the opening of the Norwegian Sea and involves dextral transpression and extension [Birkenmajer, 1981; Craddock et al., 1985].

CONCLUSIONS

1. Paleomagnetic results obtained from an extensive sample collection of Mesozoic sills and dykes in eastern Sverdrup Basin yielded a mean magnetic direction of 274°, +79° (k=173, α_{95}=3.7°, N=10), with a corresponding pole position at 157°W and 67°N (k=61, A_{95}=6.6°, N=10).

2. The application of the paleomagnetic fold test shows that the emplacement of sills and dykes preceded a major deformation characterized by tilting of strata around presently observed strike, i.e., around roughly horizontal axes. Undoubtedly, this deformation corresponds to the Eocene to early Oligocene main compressive stage of the Eurekan orogeny.

3. The virtual geomagnetic pole position falls to the east of the apparent polar wander path. This may indicate that postintrusion Eurekan deformation at some stage included an element of rotation around steeply inclined axes, although the degree of rotation and position of axes depend critically upon the age of the intrusives. Should the intrusives prove to be Cretaceous, however, the average permitted rotation agrees with the estimated rotation of Greenland between 78 and 55 Ma due to the opening of the Labrador Sea.

4. Although the absolute age of the intrusives cannot be determined paleomagnetically, there is no evidence that sill magnetization age increases with stratigraphic depth. The link between Mesozoic basin development and magmatism is therefore tenuous, and the causes of regional magmatic activity should be sought in plate tectonic processes occurring in the Arctic Ocean.

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REFERENCES

Bailey, R. C., and H. C. Halls, Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data, J. Geophys., 54, 178-182, 1984.

Balkwill, H. R., Evolution of Sverdrup Basin, Arctic Canada, Am. Assoc. Pet. Geol. Bull., 62, 1004-1028, 1978.

Balkwill, H. R., Geology of Amund Ringnes, Cornwall, and Haig-Thomas Islands, District of Franklin, Mem. Geol. Surv. Can., 390, 76 pp., 1983.

Balkwill, H. R., and F. G. Fox, Incipient rift zone, western Sverdrup Basin, Arctic Canada, in Arctic Geology and Geophysics, Mem. 8, edited by A. F. Embry and H. R. Balkwill, pp. 171-187, Canadian Society of Petroleum Geologists, Calgary, Alberta, 1982.

Balkwill, H. R., and N. E. Haimila, K/Ar ages and significance of mafic rocks, Sabine Peninsula, Melville Island, District of Franklin, Geol. Surv. Can. Pap. 78-1C, 35-38, 1978.

Birkenmajer, K., The geology of Svalbard, the western part of the Barents Sea, and the continental margin of Scandinavia, in The Ocean Basins and Margins, vol. 5, The Arctic Ocean, edited by A. E. M. Nairn et al., pp. 265-329, Plenum, New York, 1981.

Blackadar, R. G., Basic intrusions of the Queen Elizabeth Islands, District of Franklin, Geol. Surv. Can. Bull., 97, 36 pp., 1964.

Craddock, C., E. C. Hauser, H. D. Maher,

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A. Y. Sun, and Z. Guo-Qiang, Tectonic evolution of the west Spitsbergen fold belt, Tectonophysics, 114, 193-211, 1985.

Dawes, P. R., and J. W. Kerr (Eds.), Nares Strait and the Drift of Greenland: A Conflict in Plate Tectonics, Geosci., vol. 8, 392 pp., Meddelelser om Grønland, 1982.

Fricker, P. E., Geology of the Expedition area, western central Axel Heiberg Island, Canadian Arctic Archipelago, Axel Heiberg Island Res. Rep., Geol. 1, 156 pp., McGill Univ., Montreal, Canada, 1963.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, J. Geophys. Res., 54, 131-167, 1949 .

Halls, H. C., Paleomagnetism, structure, and longitudinal correlation of Middle Precambrian dykes from northwestern Ontario and Minnesota, Can. J. Earth Sci., 23, 142-157, 1986.

Irving, E., and G. A. Irving, Apparent polar wander paths, Carboniferous through Cenozoic and the assembly of Gondwana, Geophys. Surv., 5, 141-188, 1982.

Irving, E., and D. F. Strong, Palaeomagnetism of the Early Carboniferous Deer Lake Group, western Newfoundland: No evidence for mid-Carboniferous displacement of "Acadia," Earth Planet. Sci. Lett., 69, 379-390, 1984.

Jackson, K. C., and H. C. Halls, The age of the North Mokka Anticline, Axel Heiberg Island, Canadian Arctic Archipelago: An application of the paleomagnetic fold-test, Bull. Can. Pet. Geol., 33, 227-235, 1985.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, Geophys. J. R. Astron. Soc., 62, 669-718, 1980.

LaBreque, J. L., D. V. Kent, and S. C. Cande, Revised magnetic polarity time scale for Late Cretaceous and Cenozoic time, Geology, 5, 330-335, 1977.

Larochelle, A., and R. F. Black, An application of palaeomagnetism in estimating the age of rocks, Nature, 198, 1260-1262, 1963.

Larochelle, A., R. F. Black, and R. K. Wanless, Palaeomagnetism of the Isachsen Diabasic rocks, Nature, 208, 179, 1965.

Larson, R. L., and T. W. C. Hilde, A revised time scale of magnetic reversals for the Early Cretaceous and Late Jurassic, J. Geophys. Res., 80, 2586-2594, 1975.

May, S. R., and R. F. Butler, North American Jurassic apparent polar wander: Implications for plate motion, paleogeography and Cordilleran tectonics, J. Geophys. Res., 91, 11,519-11,544, 1986.

McDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in paleomagnetic studies, J. Geophys. Res., 85, 3659-3669, 1980.

McElhinny, M. W., Statistical significance of the fold test in palaeomagnetism, Geophys. J. R. Astron. Soc., 8, 338-340, 1964.

McFadden, P. L., and F. J. Lowes, The discrimination of mean directions drawn from Fisher distributions, Geophys. J. R. Astron. Soc., 67, 19-31, 1981.

Miall, A. D., The Nares Strait problem: A re-evaluation of the geological evidence in terms of a diffuse, oblique-slip plate boundary between Greenland and the Canadian Arctic Islands, Tectonophysics, 100, 227-239, 1983.

Miall, A. D., Sedimentation and tectonics of a diffuse plate boundary: The Canadian Arctic Islands from 80 Ma B.P. to the present, Tectonophysics, 107, 261-277, 1984.

Nicholson, C., L. Seeber, P. Williams, and L. R. Sykes, Seismic evidence for conjugate slip and block rotation within the San Andreas Fault system, Southern California, Tectonics, 5, 629-648, 1986.

Osadetz, K., and B. Ricketts, Mesozoic volcanic rocks of Sverdrup Basin (abstract), paper presented at the Geological Association of Canada meeting, p. A45, Fredericton, Canada, 1985.

Peirce, J. W., The evolution of the Nares Strait lineament and its relation to the Eurekan orogeny, in Nares Strait and the Drift of Greenland: A Conflict in Plate Tectonics, Geosci., vol. 8, edited by P. R. Dawes and J. W. Kerr, pp. 237-252, Meddelelser om Grønland, 1982.

Srivastava, S. P., Evolution of the Eurasian Basin and its implication to the motion of Greenland along Nares Strait, Tectonophysics, 114, 29-53, 1985.

Steiner, M. B., Mesozoic apparent polar wander and plate motions of North America, in Mesozoic Paleogeography of West Central United States, edited by M.

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W. Reynolds and E. D. Dolly, pp. 1-11, Rocky Mountain Section, Society of Economic Paleontologists and Mineralogists, Denver, Colo., 1983.

Thorsteinsson, R., and E. T. Tozer, Geology of the Arctic Archipelago, Geol. and Econ. Min. of Can., Econ. Geol. Rep., edited by R. J. W. Douglas, pp. 458-590, Geological Survey of Canada, Ottawa, 1970.

Torsvik, T. H., R. Løvlie, and B. A. Sturt, Palaeomagnetic argument for a stationary Spitsbergen relative to the British Isles (Western Europe) since late Devonian and its bearing on North Atlantic reconstruction, Earth Planet. Sci. Lett., 75, 278-288, 1985.

van Berkel, J. T., H. Hugon, W. M. Schwerdtner, and J. L. Bouchez, Study of anticlines, faults and diapirs in the central Eureka Sound Fold Belt, Canadian Arctic Islands: Preliminary results, Bull. Can. Pet. Geol., 31, 109-116, 1983.

Vincenz, S. A., and M. Jelenska, Paleomagnetic investigations of Mesozoic and Paleozoic rocks from Svalbard, Tectonophysics, 114, 163-180, 1985.

Watson, G. S., Analysis of dispersion on a sphere, Mon. Not. R. Astron. Soc., Geophys. Suppl., 7, 153-159, 1956.

Watts, D. R., Palaeomagnetism of the Lower Carboniferous Billefjorden Group, Spitsbergen, Geol. Mag., 122, 383-388, 1985.

Williamson, M. C., and N. A. Van Wagoner, Volcanism at the Alpha Ridge and in the northwest Sverdrup Basin: Is there a case for a common origin? (abstract), paper presented at Geological Association of Canada meeting, p. A69, Fredericton, Canada, 1985.

Wynne, P. J., E. Irving, and K. Osadetz, Paleomagnetism of the Esayoo Formation (Permian) of northern Ellesmere Island: Possible clue to the solution of the Nares Strait dilemma, Tectonophysics, 100, 241-256, 1983.

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